

UNIT - III* Phase velocity & Group velocity

$$y = A \sin(kx - \omega t)$$

$$kx - \omega t = \text{const.}$$

$$kx = \omega t$$

$$\boxed{v_{\text{phase}} = \frac{dx}{dt} = \frac{\omega}{k}} \rightarrow \text{Phase velocity}$$

$$\boxed{\frac{dx}{dt} = \frac{\Delta \omega}{\Delta k} = v_{\text{group}}} \rightarrow \text{Group velocity}$$

* Depth of penetration / skin depth

$$\boxed{\delta = \frac{1}{\alpha}}$$

$$\alpha = \sqrt{\frac{\omega \mu \sigma}{2}}$$

$$\therefore \boxed{\delta = \sqrt{\frac{2}{\omega \mu \sigma}}}$$

* Conductors & dielectrics (Pg 332)

$\frac{\sigma}{\omega \epsilon} =$ the ratio of conduction current density to displacement current density.

Good conductors $\rightarrow \frac{\sigma}{\omega \epsilon} \gg 1$ over entire radio frequency range.

Good dielectrics $\rightarrow \frac{\sigma}{\omega \epsilon} \ll 1$ in the radio frequency range.

$\frac{\sigma}{\omega \epsilon} = \text{loss tangent} / \text{dissipation factor}$

* Impedance of Conducting Medium

$$\frac{E}{H} = \eta = \sqrt{\frac{j\omega\mu}{\sigma + j\omega\epsilon}}$$

① Lossless medium $\rightarrow (\sigma = 0)$

$$\eta = \sqrt{\frac{\mu}{\epsilon}} = \sqrt{\frac{\mu_0}{\epsilon_0}} = 120\pi \text{ or } 377 \Omega$$

② Lossy medium $\rightarrow (\sigma \gg 1)$

$$\eta = \sqrt{\frac{j\omega\mu}{\sigma}} = \sqrt{\frac{\omega\mu}{\sigma}} < 45^\circ$$

Polarization

- ① linear
- ② circular
- ③ elliptical

$$x = E_x = E_{mx} \sin \omega t$$

$$y = E_y = E_{my} \sin (\omega t + \phi)$$

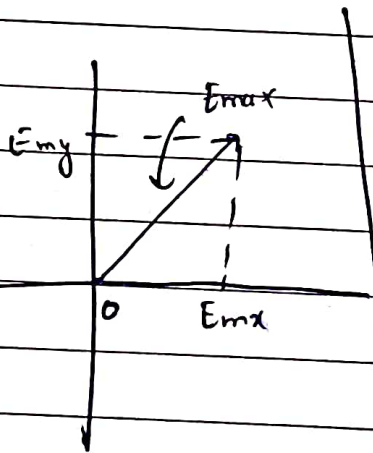
} Linear

$$x^2 + y^2 = E_m^2 \quad \text{Circular}$$

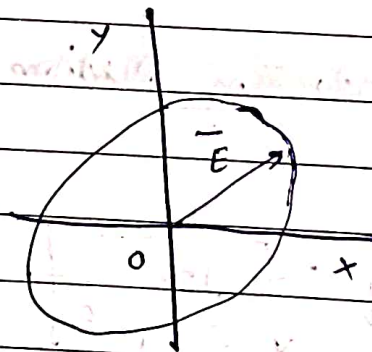
$$x = E_x = E_m \sin \omega t$$

$$y = E_y = E_m \sin (\omega t + \pi/2) = E_m \cos \omega t$$

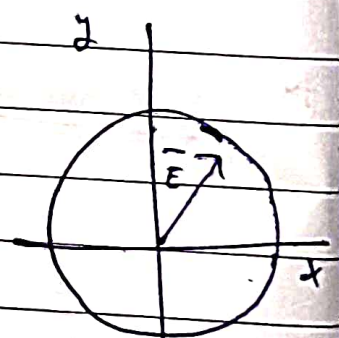
} Elliptical



linear



elliptical



Circular

$$E_r = E_{r0} e^{jK_1 z}$$

$$H_r = H_{r0} e^{jK_1 z}$$

$$H_r = -\frac{E_{r0}}{n_1} e^{jK_1 z}$$

$$\text{At } z = 0$$

$$E_i + E_r = E_t$$

$$H_i + H_r = H_t$$

$$\frac{E_{i0}}{n_1} - \frac{E_{r0}}{n_1} = \frac{E_{t0}}{n_2}$$

$$\frac{E_i}{n_1} - \frac{E_r}{n_1} = \frac{E_t}{n_2}$$

$$\frac{E_i}{n_1} - \frac{E_r}{n_1} = \frac{E_i + E_r}{n_2}$$

$$(E_t = E_i + E_r)$$

$$\boxed{\frac{E_r}{E_i} = \frac{n_2 - n_1}{n_2 + n_1} = \Gamma}$$

(Reflection coefficient)

Again,

$$\frac{E_{i0}}{n_1} - \frac{E_{r0}}{n_1} = \frac{E_{t0}}{n_2}$$

$$\frac{E_i}{n_1} - \frac{(E_t - E_i)}{n_1} = \frac{E_t}{n_2}$$

$$(E_r = E_t - E_i)$$

$$\boxed{\frac{E_t}{E_i} = \frac{2n_2}{n_2 + n_1} = T}$$

(Transmission coefficient)

UNIT-IV

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* Transit time effect

$$t_r = \frac{l}{v}$$

voltage at BB' $\rightarrow V_p$

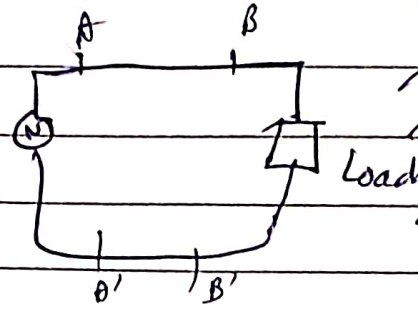
" " AA' $\rightarrow V_q$

$$T \gg t_r$$

$$T \gg \frac{l}{v}$$

velocity $\frac{1}{f} \gg \frac{l}{v}$

frequency $\frac{v}{l} \gg l \Rightarrow \lambda \gg l$
wavelength



$T \rightarrow$ time period of signal

* Primary constants of transmission line

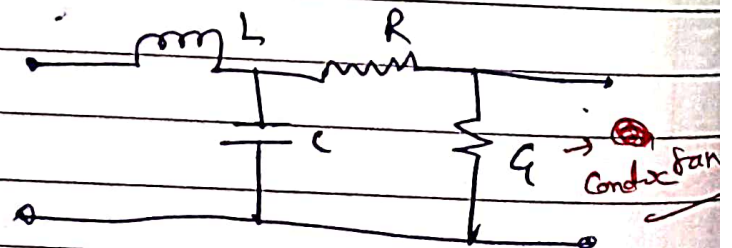
- Its for small part of line

$$L \rightarrow \text{H/m}$$

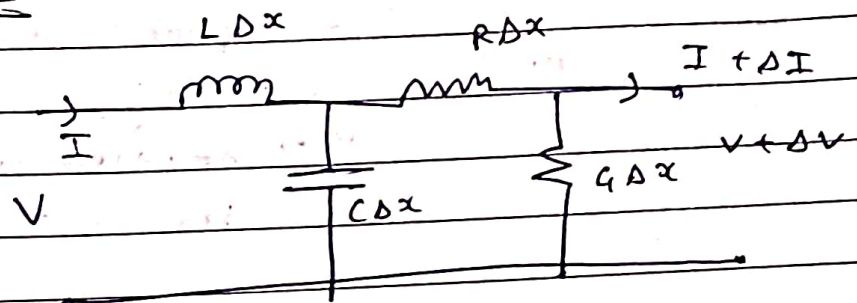
$$C \rightarrow \text{F/m}$$

$$R \rightarrow \Omega/\text{m}$$

$$G \rightarrow \text{S/m}$$



Transmission line equation



$$\Delta V = -(R\Delta x + j\omega L\Delta x) I$$

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta V}{\Delta x} = -(R + j\omega L) I$$

$$\frac{dV}{dx} = -(R + j\omega L) I \quad \text{--- (1)}$$

$$\Delta I = -(G\Delta x + j\omega C\Delta x) V$$

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta I}{\Delta x} = -(G + j\omega C) V$$

$$\frac{dI}{dx} = -(G + j\omega C) V \quad \text{--- (2)}$$

Diff. eqn (1)

$$\frac{d^2 V}{dx^2} = -(R + j\omega L) \frac{dI}{dx}$$

$$\frac{d^2 V}{dx^2} = (R + j\omega L)(G + j\omega C) V$$

$$\boxed{\frac{d^2 V}{dx^2} = \gamma^2 V}$$

$\gamma^2 \rightarrow (\text{gamma})^2$
Primary constants

$$V(x,t) = (V^+ e^{-\gamma x} + V^- e^{+\gamma x}) \cdot e^{j\omega t}$$

Gamma $\rightarrow$ Propagation constant

$$\gamma = \alpha + j\beta$$

$\alpha \rightarrow$ Attenuation const.

$\beta \rightarrow$ Phase const.

In loss less medium ($\alpha = 0$)

$$V(x,t) = V^+ e^{-(\alpha + j\beta)x} e^{j\omega t} + V^- e^{+(\alpha + j\beta)x} e^{j\omega t}$$

$$V(x,t) = V^+ e^{-j\beta x} e^{j\omega t} + V^- e^{j\beta x} e^{j\omega t}$$

$$V(x,t) = V^+ e^{j(\omega t - \beta x)} + V^- e^{-j(\omega t + \beta x)}$$

$$V(x,t) = V^+ \cos(\omega t - \beta x) + V^- \cos(\omega t + \beta x)$$

travelling
+x dir

travelling
-x dir

• Same is for $I(x,t)$ also, the eqn will travel in + & - direction

• So from this we conclude V & I has the wave nature of travelling in opposite directions.

* Characteristic Impedance

$$\frac{dV}{dx} = -(R + j\omega L) I$$

$$\frac{d}{dx} (V^+ e^{-\gamma x} + V^- e^{+\gamma x}) = -(R + j\omega L) \{ I^+ e^{-\gamma x} + I^- e^{+\gamma x} \}$$

$$V^+ e^{-\gamma x} \times (-\gamma) + V^- e^{+\gamma x} \times (\gamma) = -(R + j\omega L) \{ I^+ e^{-\gamma x} + I^- e^{+\gamma x} \}$$

γ Gamma

Part of eqⁿ travelling forward.

$$V^+ e^{-\gamma x} \times (-\gamma) = -(R + j\omega L) \{ I^+ e^{-\gamma x} \}$$

$$\frac{V^+}{I^+} = \frac{R + j\omega L}{\gamma}$$

γ Gamma

$$\left\{ \begin{array}{l} \gamma^2 = (R + j\omega L)(G + j\omega C) \\ \gamma \text{ Gamma} \end{array} \right\}$$

$$Z_0 = \frac{V^+}{I^+} = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \rightarrow \text{Characteristic Impedance}$$

$\frac{V}{I} = \text{impedance, } \Omega$
 R, L, G, C are primary
 characters. So
 characteristic impedance

Part of eqⁿ travelling backward.

$$V^- e^{+\gamma x} \times \gamma = -(R + j\omega L) I^- e^{+\gamma x}$$

$$\frac{V^-}{I^-} = -\frac{(R + j\omega L)}{\gamma}$$

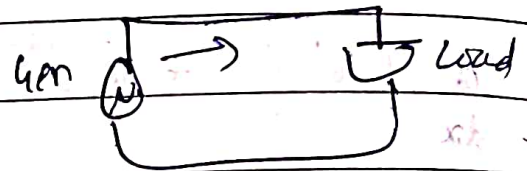
$$Z_0 = \frac{V^-}{I^-} = -\sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

(-ve sign implies that it is receiving the energy)

* Different types of transmission line

① Lossless Tx line :

$$R = 0, G = 0$$



Gamma $\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$

$$\gamma = \sqrt{j^2 \omega^2 LC} = j\omega \sqrt{LC}$$

$$\alpha + j\beta = j\omega \sqrt{LC}$$

compare

$$\alpha = 0,$$

$$\beta = \omega \sqrt{LC} \rightarrow \frac{2\pi}{\lambda} = 2\pi f \sqrt{LC}$$

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} = \sqrt{\frac{L}{C}}$$

$$f_d = \frac{1}{\sqrt{LC}}$$

$$V_P = \frac{1}{\sqrt{LC}}$$

$$f = \frac{v}{\lambda}$$

② Low Loss Tx line

$$R \ll \omega L, \quad G \ll \omega C$$

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$\gamma = \sqrt{j\omega L \left(1 + \frac{R}{j\omega L}\right) j\omega C \left(1 + \frac{G}{j\omega C}\right)}$$

$$= \left\{ j\omega L \left(1 - \frac{jR}{\omega L}\right) j\omega C \left(1 - \frac{jG}{\omega C}\right) \right\}^{1/2}$$

$$= j\omega \sqrt{LC} \left\{ \left(1 - \frac{jR}{\omega L}\right) \left(1 - \frac{jG}{\omega C}\right) \right\}^{1/2}$$

$$= j\omega \sqrt{LC} \left\{ 1 - \frac{jR}{\omega L} - \frac{jG}{\omega C} \right\}^{1/2}$$

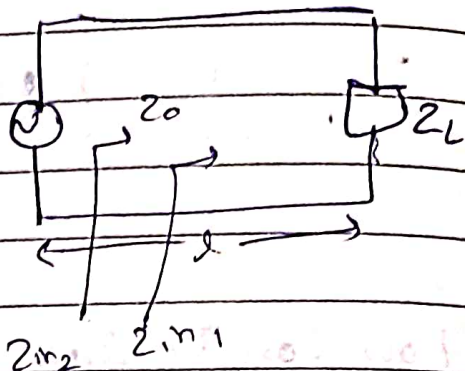
$$\gamma = \cancel{j\omega \sqrt{LC}} + \frac{R}{2} \sqrt{\frac{C}{L}} + \frac{G}{2} \sqrt{\frac{L}{C}}$$

$$\alpha + j\beta = \underbrace{j\omega \sqrt{LC}}_{\beta} + \underbrace{\frac{R}{2} \sqrt{\frac{C}{L}} + \frac{G}{2} \sqrt{\frac{L}{C}}}_{\alpha}$$

$$Z_0 = \sqrt{\frac{L}{C}} \left\{ 1 - \frac{jR}{2\omega L} + \frac{jG}{2\omega C} \right\}$$

* Input Impedance of transmission line

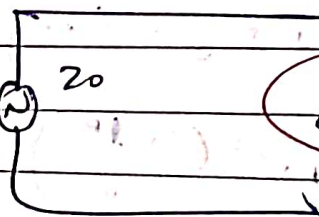
① Loss less Medium :



$$Z_{in} = Z_0 \left[\frac{Z_L + j Z_0 \tan \beta l}{Z_0 + j Z_L \tan \beta l} \right]$$

* Input Impedance of open ckt

$$Z_i = Z_0 \left[\frac{Z_L + j Z_0 \tan \beta l}{Z_0 + j Z_L \tan \beta l} \right]$$



open ckt

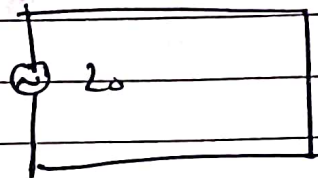
$$Z_L = \infty$$

$$Z_i = Z_0 \times \cancel{Z_L} \left\{ \frac{1 + j \cancel{\frac{Z_0}{Z_L}} \tan \beta l}{\cancel{\frac{Z_0}{Z_L}} + j \tan \beta l} \right\}$$

$$Z_{in} = -j Z_0 \cot \beta l$$

* Input Impedance of short circuit

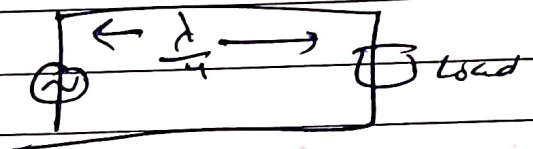
$$Z_{in} = Z_0 \left[\frac{j Z_0 \tan \beta l}{Z_0} \right]$$



short circuit
 $Z_L = 0$

$$Z_{in} = j Z_0 \tan \beta l$$

$$Z_{in} = \frac{Z_0^2}{Z_L} \rightarrow \text{for } \lambda/4 \text{ long on}$$



* Distortionless Tx line

$$\frac{R}{L} = \frac{G}{C}$$

$$\gamma = \sqrt{(R + j\omega L)(G + j\omega C)}$$

$$= \sqrt{R \left(1 + j\omega \frac{L}{R}\right) G \left(1 + j\omega \frac{C}{G}\right)}$$

$$= \sqrt{RG \left(1 + j\omega \frac{C}{G}\right)^2}$$

$$\alpha + j\beta = \sqrt{RG} \left(1 + j\omega \frac{C}{G}\right)$$

$$\alpha = \sqrt{RG}, \quad \beta = j\sqrt{RG} \omega \frac{C}{G}$$

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}}$$

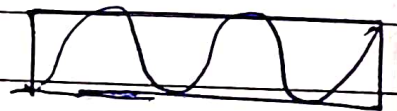
$$Z_0 = \sqrt{\frac{R}{G}} \quad \text{or} \quad Z_0 = \sqrt{\frac{L}{C}}$$

* VSWR

→ Voltage standing wave ratio

$$VSWR = \frac{V_{\max}}{V_{\min}} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

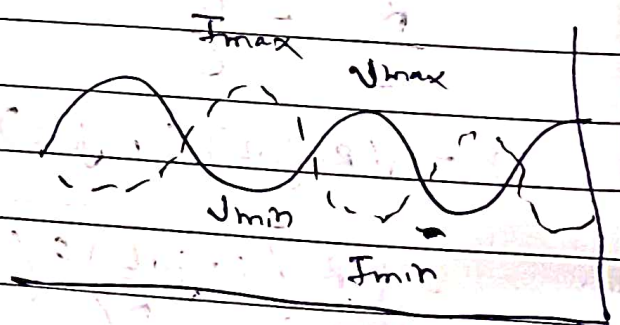
$$V_{\max} = Z_{\min} = \frac{-1}{2\beta} (\phi + (2m+1)\pi)$$



$\{m=0, 1, 2, \dots\}$

$$V_{\min} = Z_{\max} = \frac{-1}{2\beta} (\phi + 2m\pi)$$

$$Z_{\max} = \frac{V_{\max}}{I_{\min}} = Z_0 S$$



$$Z_{\min} = \frac{V_{\min}}{I_{\max}} = \frac{Z_0}{S}$$

← $\frac{\lambda}{2}$

* Range of Γ & VSWR (S)

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

for matched condition

$$Z_L = Z_0$$

$$\Gamma = 0$$

$$\Gamma = 1$$

$$0 < \Gamma \leq 1$$

$$VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

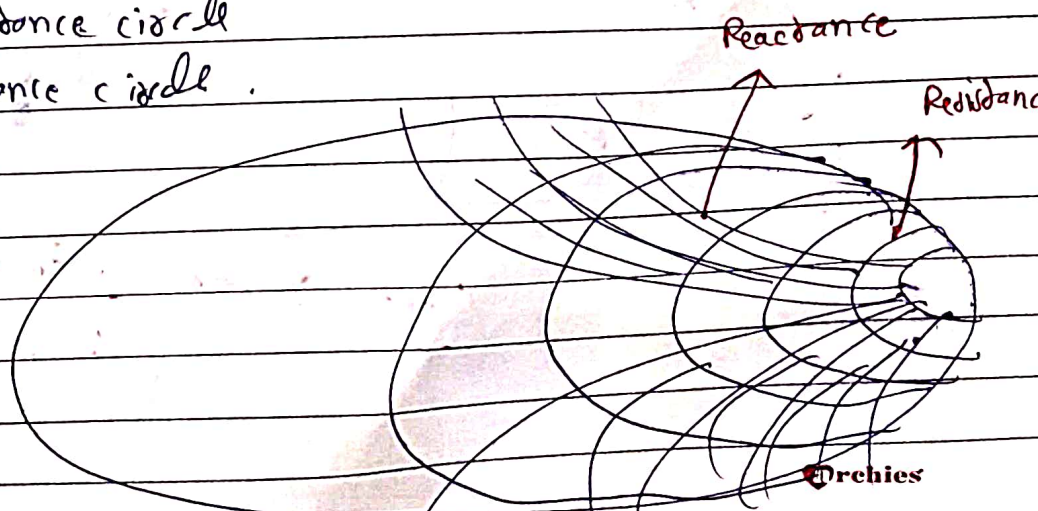
$$|VSWR|_{\Gamma=0} = 1$$

$$|VSWR|_{\Gamma=1} = \infty$$

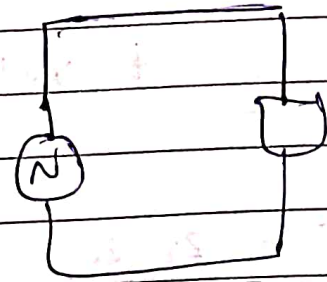
$$1 \leq VSWR < \infty$$

* Smith Chart

- Constant Resistance circle
- Constant Reactance circle



eg.



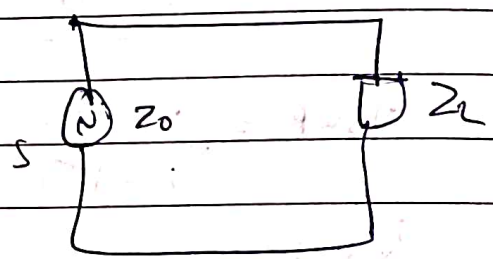
$Z_L = 50 + j100$

Now we will plot $1 + j2$

$Z_{LN} = \frac{Z_L}{Z_0}$

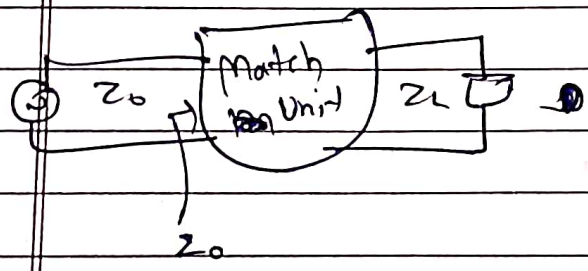
$= \frac{50 + j100}{50} = 1 + j2$

* Impedance Matching



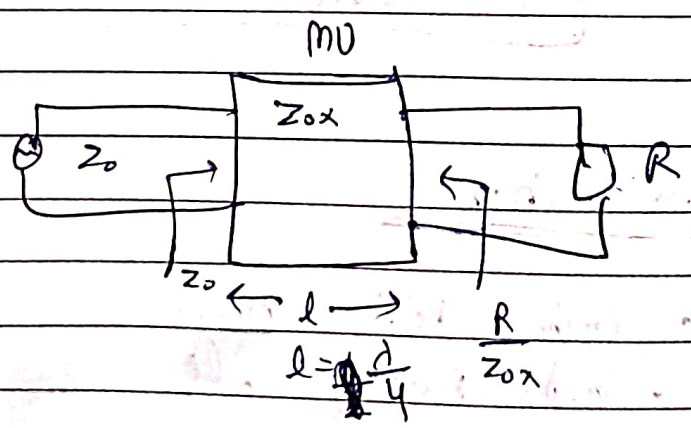
if $Z_L = Z_0$

if $Z_L \neq Z_0$



① Resistive Load :

$Z_{in} = \frac{Z_{ox}^2}{R}$



$Z_{ox} = \sqrt{R Z_0}$

Also Quarter wave transformer
!! $l = \frac{d}{4}$

